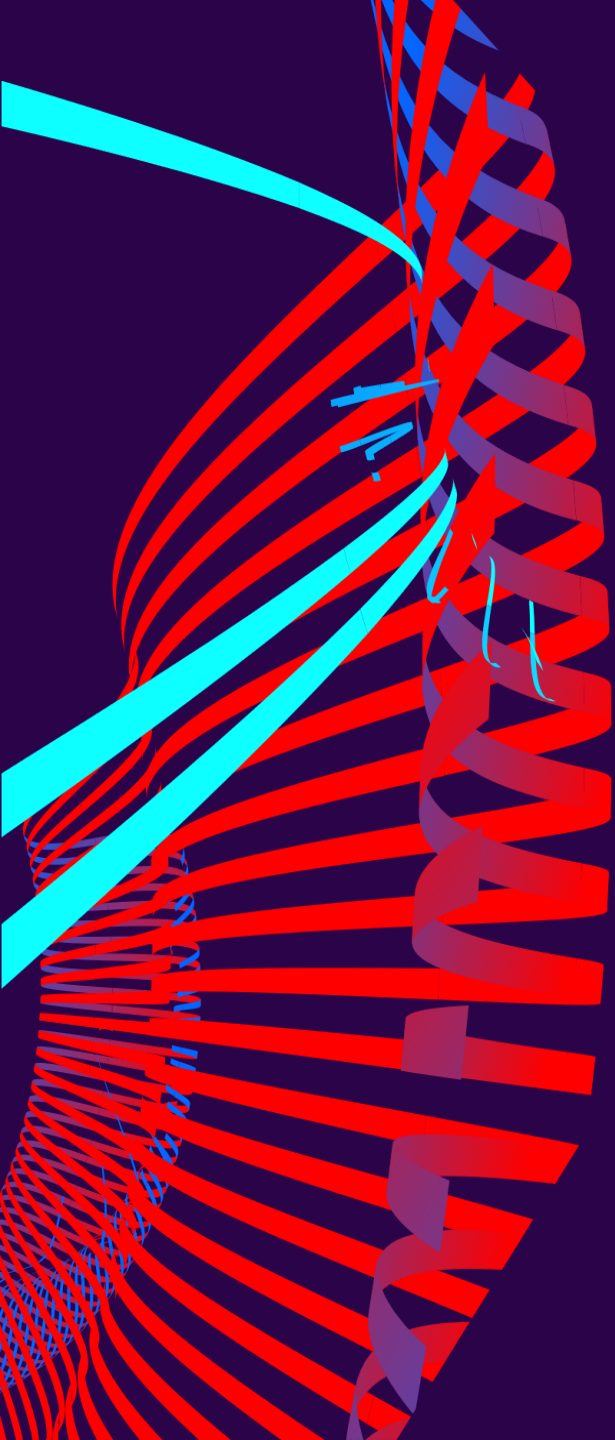




JSS- 1 BASIC SCIENCE

WEEK 1: 1ST TERM

TOPIC: INTRODUCTION TO SCIENCE

LEADERS TUTORS

SUB TOPICS

- 1) Concept of Science and Scientific Process.**
- 2) Science Related Occupations.**
- 3) Benefits of Science to mankind.**
- 4) Applications of Science and Scientific Inventions.**

WHAT STUDENTS WILL GAIN AT THE END OF THE LESSON

At the end of the lesson, students should be able to:

- 1. Explain the concept of Science and Scientific Process.**
- 2. Analyze Science related occupations.**
- 3. Mention the benefits of science to mankind.**
- 4. Highlight the applications of Science and Scientific inventions.**

(1A). CONCEPTS OF SCIENCE AND SCIENTIFIC PROCESSES.

Science is a systematic study of the nature and manners of an object and the natural universe that is established around measurement, experiment, observation and formulation of laws.

It is a broad career field with many specialties and areas of study.

Science describes an area of knowledge, typically about something in the physical world that can be explained in terms of scientific observation or the scientific method.

(1B) CONCEPTS OF SCIENCE AND SCIENTIFIC PROCESSES.

The scientific process is a discovery process that has evolved over several hundred years and can be summarized as follows:

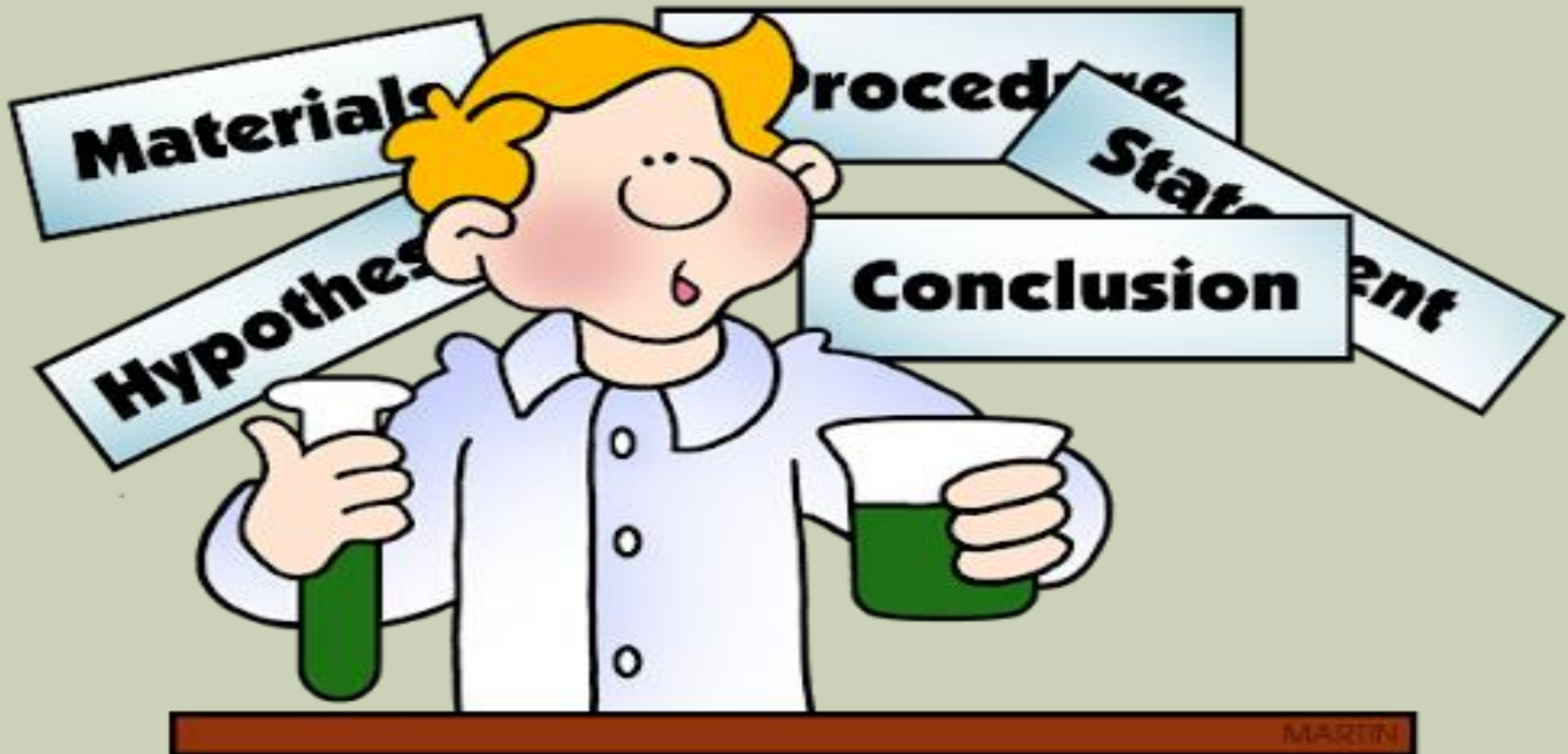
A phenomenon in the physical world is observed.

An explanation, or hypothesis, for the phenomenon is formed.

The hypothesis is tested by means of objective, reproducible experiments.

(1C) CONCEPTS OF SCIENCE AND SCIENTIFIC PROCESSES.

CONCEPT OF SCIENCE



(2A) SCIENCE RELATED OCCUPATIONS

- **Biologist**
- **Chemist**
- **Physicist**
- **Astronomer**
- **Geologist**
- **Meteorologist**
- **Medical Laboratory Scientist**

(2B)SCIENCE RELATED OCCUPATIONS.

- **Data Scientist**
- **Genetic Engineers**
- **Microbiologist**
- **Neuroscientist**
- **Biochemist**
- **Environmental Scientist**

(2C)SCIENCE RELATED OCCUPATIONS.

Do you recognize any profession here?

Here we have: a pilot, a medical laboratory scientist, a data scientist, a medical doctor, a computer programmer, a nutritionist, a neuroscientist, a civil engineer and a biochemist.



PICTURE OF MEDICAL LABORATORY SCIENTISTS IN THE LABORATORY.

A medical laboratory scientist (MLS) also known as a medical technologist or a clinical laboratory scientist work to analyse a variety of biological specimens.



GENETIC ENGINEER

Genetic engineers experiment with altering, splicing, eliminating, and rearranging genes to obtain specific results.



DATA SCIENTIST

Data scientists use data to understand and explain the phenomena around them, and help organizations make better decisions.



BIOCHEMIST

Biochemists are sometimes called molecular biologists or cellular biologists. They study the molecular mechanisms by which cells feed, divide, and grow.



BENEFITS OF SCIENCE TO MANKIND.

- **It contributes to ensuring a longer and healthier life.**
- **It helps to monitor our health**
- **It provides medicine to cure our diseases**
- **It alleviates aches and pains**
- **It helps to provide water for our basic needs.**
- **It aids communication with better connectivity.**
- **It helps to organize our daily activities**
- **It helps our work to be done faster and makes life easier**
- **It helps us to know more about other cultures and societies.**

(3A). APPLICATIONS OF SCIENCE AND SCIENTIFIC INVENTIONS

- There are four major branches of science are, Mathematics and logic, biological science, physical science and social science, each branch is categorized in different type of subjects that covers different areas of studies such as chemistry, physics, math, biology etc.

SECTIONS OF SCIENCE

- **Biology:** the study of living things. Biological science is divided into different sub topics. One of them is Zoology. It is a category under BIOLOGY
- **CHEMISTRY:** is the study of different substances, the changes they undergo and their compositions. OGY that focuses on the study of animal life.
- **PHYSICS:** is the study of how matter and energy interact. It encompasses the study of matter, its motion and behavior along with energy and force.

(3B) APPLICATION TO SCIENCE AND SCIENTIFIC INVENTIONS.

- **Biophysics**
- **Mechanics**
- **Acoustics**
- **Optics**
- **Thermodynamics**
- **Astrophysics**

GENERAL LABORATORY APPARATUS

LABORATORY APPARATUS



Pipette



Test tube rack



Test tube



Erlenmeyer flask



Beaker



Bunsen burner



Barometer



Indicator



Stopwatch



Speedometer



Protractor



Alcohol burner



Syringe



Graduated cylinder



Dropper



Tongs



Tuning fork



Stethoscope



Magnet



Magnifiers



Telescope



Microscope



Thermometer



Friability tester



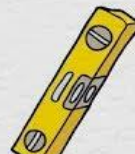
Pulleys



Tape



Dry-cell battery



Level



Balance scale



Charts



Dissecting



Earth science

ASSIGNMENT

- 1. List ten science related occupations.**
- 2. Why is Science important?**

TOPIC QUESTION

- 1) What is Science?**
- 2) What are scientific processes?**
- 3) List ten science related occupations.**
- 4) Mention five benefits of science to mankind**
- 5) Highlight the applications of science and scientific inventions.**

REVIEW OF THE LESSON

At the end of the lesson, we were able to:

- 1. Define the concept of Science.**
- 2. Explain the scientific process.**
- 3. List science related occupations.**
- 4. Mention benefits of science to mankind.**
- 5. Highlight the applications of science and scientific inventions.**

THANK YOU VERY MUCH

Remember

There are great opportunities for you when you study well.

Also, after answering questions on any topic in Basic Science, try and master the real life application.

I see a very bright future ahead of you.

